MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 1613

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, RF power amplifier

Outline	---	8-6 (EIA) Ⓐ
Base	---	B7-22
Envelope	---	MT8
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	7	8
Element	---	sh	h	p	g2	g1	h	k, g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ic1	Pi	Pg2	Pp	Ⓐ Ehk	Alt
Unit:	V	Vdc	Vdc	Vdc	mAdc	mAdc	W	W	W	v	ft
C Telep:											
Maximum:	6.9	275	-100	275	50	5	11.5	2.0	7	100	10,000
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---
C Teleg:											
Maximum:	6.9	350	-100	275	50	5	17.5	2.5	10	100	10,000
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	250	-16.5	250	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See note 1	0.65	II	Ic1	0	-3.0	μAdc
1256	*Electrode current (anode)*		*0.65*	*II*	*Ib*	25	43	mAdc
1236	Power oscillation (1)	Ib = 50 mAdc; Ic1 = 4 to 6 mAdc; t = 120; Eb = 350 Vdc; Ec2 = 200 Vdc; Rg1 = 10,000 *ohms*; F = 45 MHz (see note 1)	0.65	II	Po	7.0	---	W
1231	Emission	Ec1 = Ec2 = Eb = 50 Vdc	0.65	II	Is	175	---	mAdc
Ⓐ 1201	Short and discontinuity detection		0.4	II	---	---	---	---
Ⓐ Appendix D, 30(a), 40	Visual and mechanical inspection criteria		---	---	---	---	---	---
Ⓐ Appendix D, 30(b)	Dimensions	Outline 8-6; dimensions A, B, C, D, L, and M	1.0	I	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		6.5	S3	If	640	760	mA
Ⓐ 1336	Heater-cathode leakage		6.5	S3	Ihk	---	75	μAdc
1341	Power output	Esig = 11.6 Vac; Rp = 7,000 ohms	6.5	S3	Po	2.75	---	W
1306	Transconductance	Ib = 31 mAdc	6.5	S3	Sm	2,075	2,925	μmhos
1256	Electrode current (screen grid)		6.5	S3	Ic2	4.0	8.0	mAdc
1331	Direct-interelectrode capacitance		6.5	Code E	Cgp Cin Cout	0.09 4.5 9.5	0.6 8.5 17.5	pF pF pF
1031	*Low-frequency vibration*	Ec1 = -27 Vdc; Rp = 2,000 ohms	6.5	S3	Ep	---	300	mVac
Ⓐ 1101	Secureness of base, cap. or insert		6.5	S3	---	---	---	---
Ⓐ 1111	Base pin solder depth		6.5	S3	---	---	---	---
Ⓐ 1211	Insulation of electrodes		4.0	S3	R	10	---	Meg
Ⓐ 1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
1501	Life-test provisions (intermittent) (500 hours)	Group A; Eb = Ec2 = 300 Vdc; Ec1 = -20 Vdc; Rg1 = 0.1 Meg; Ehk = 100 V	---	---	---	---	---	---
---	Life-test end points (intermittent) (500 hours)	Power output or Transconductance	--- ---	--- ---	Po Sm	1.7 1,625	--- ---	W μmhos
	Periodic-check test							
1236	Power oscillation (2)	Ib = 50 mAdc; Ic1 = 4 to 6 mAdc; t = 120; Eb = 350 Vdc; Ec2 = 200 Vdc; Rg1 = 10,000 ohms; F = 8 MHz	---	---	Po	8.5	---	W

NOTE:

1. This test to be performed at the conclusion of the holding period.

The margins of this tube specification sheet are marked with a revision letter (A) to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Air Force - 11

(Project 5960-2199-57)

☆ U. S. GOVERNMENT PRINTING OFFICE: 1968-341-510/A-1057

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